

Scalable Data Engineering and Relational Pipeline Architecture for Banking Analytics: An End-to-End Implementation Using Python, MySQL, and Power BI

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ABSTRACT — Large-scale institutional bank consolidations generate complex heterogeneous datasets characterized by divergent reporting schemas, data entry anomalies, inconsistent fiscal identifiers, and legacy storage formats. Analyzing such structural transformations requires an integrated data engineering pipeline capable of synthesizing micro-data, executing cleaning and imputation protocols, enforcing relational consistency, and surfacing multi-dimensional aggregations through business intelligence dashboards. This paper details the software architecture, design patterns, and implementation strategies of an end-to-end financial analytics pipeline constructed to evaluate the 2017 State Bank of India (SBI) consolidation. The tech stack integrates: (i) an algorithmic data synthesis engine in Python using NumPy and Pandas; (ii) an automated Extract, Transform, Load (ETL) pipeline using SQLAlchemy and PyMySQL; (iii) a relational database engine hosting optimized analytical aggregations; and (iv) an analytical dashboard model deployed via Microsoft Power BI (.pbix). We describe data sanitation techniques, group-median imputation algorithms for missing risk metrics, schema design strategies, and DAX analytical models. The architecture proves that lightweight, open-source data engineering stacks can reliably support high-throughput econometric research and enterprise reporting.

Keywords: Data Engineering, ETL Pipeline, SQLAlchemy, MySQL Architecture, Power BI, Business Intelligence, Data Imputation, Software Design Patterns.

I. ARCHITECTURAL OVERVIEW & SYSTEM DESIGN

Evaluating structural banking consolidations presents distinct systems-engineering challenges. Financial data originating from legacy systems often contains unstripped formatting characters, inconsistent date serializations, and systemic missingness caused by misaligned reporting cutoffs across merged subsidiaries. To address these hurdles, we implemented a modular, four-layer data processing pipeline, illustrated in Figure 1.



A. Pipeline Modularization and Responsibilities

- **Synthesis & Injection Engine (`generate_data.py`):** Simulates 4,000 branch observations across 8 fiscal years with defined stochastic parameters reflecting pre-merger vulnerabilities and post-merger synergies, while injecting realistic dirty-data artifacts.
- **ETL Ingestion Engine (`db_connection.py`):** Implements data ingestion, format normalization, entity whitespace trimming, mixed-format date parsing, and robust median imputation.
- **Relational Storage (`sbi_merger_analytics_db`):** Manages persistence, enforces column constraints, and runs analytical SQL grouping queries.
- **Semantic BI Layer (`merger_dashboard.pbix`):** Ingests the cleaned schema to drive visual models and interactive spatial performance diagnostics.

II. COMPONENT TECHNOLOGIES & ARCHITECTURE SPECS

Table I summarizes the core software packages, their versions, and their functional scope across the project lifecycle.

TABLE I: Technical Stack Specification & Component Responsibilities

Stack Layer	Technology / Library	Operational Responsibility	Performance / Reliability Rationale
Data Generation	Python 3.10+ / NumPy	Stochastic normal distributions, seed reproducibility	Vectorized array sampling for 4,000 observation instances
Data Sanitization	Pandas 2.0+	Series regex strip, numeric coercion, missingness imputation	High-throughput vectorized text and null operations
Database Gateway	SQLAlchemy / PyMySQL	Engine pooling, connection parameter URL-quoting	Production-grade connection lifecycle management
Relational Storage	MySQL Server 8.0	Table schema, transactional storage, aggregations	ACID guarantees, multi-column secondary B-tree indices
Analytical Queries	Structured Query Language (SQL)	Pre/post phase categorization, window grouping	Deterministic execution plans for spatial performance
Semantic BI Model	Microsoft Power BI Desktop	Star-schema relationships, DAX measures, visualizations	In-memory VertiPaq engine for sub-second analytical pivots

III. DATA SANITIZATION & IMPUTATION ENGINE

A. String Coercion and Character Cleansing

Banking data commonly aggregates metrics formatted as display strings (e.g., percentages containing "%" and currency fields with thousands commas ","). Feeding these into analytical pipelines causes silent type casting failures. The ingestion engine strips these characters across all targeted series using vectorized string operations:

```
# Vectorized Percentage and Currency Transformation
pct_cols = ["gross_npa_pct", "net_npa_pct", "cost_to_income_pct", "roa_pct", "casa_ratio_pct", "car_crar_pct"]
for col in pct_cols: df[col] = pd.to_numeric(df[col].astype(str).str.replace("%", "").str.strip(), errors="coerce")
curr_cols = ["advances_inr_cr", "deposits_inr_cr"]
for col in curr_cols: df[col] = pd.to_numeric(df[col].astype(str).str.replace(",", "").str.strip(), errors="coerce")
```

B. Entity-Level Group-Median Missingness Imputation

Simple global mean imputation distorts risk distributions because historical associate banks exhibited structurally divergent cost-to-income and asset quality metrics. For example, replacing a missing Net NPA value in a State Bank of Patiala branch with the global average would underestimate that branch's pre-merger stress. Consequently, the pipeline uses a partition-based group-median imputation strategy:

```
# Group-Partitioned Imputation Strategy
df["cost_to_income_pct"] = df.groupby("entity_name")["cost_to_income_pct"] \
    .transform(lambda s: s.fillna(s.median()))
df["net_npa_pct"] = df.groupby("entity_name")["net_npa_pct"] \
    .transform(lambda s: s.fillna(s.median()))
```

This ensures that missing values preserve the specific risk and efficiency profile of each individual operating bank.

C. Heterogeneous Date Format Harmonization

Merger-cutoff timestamps often arrive with conflicting date serializations (e.g., 01/04/2017 versus 2017-04-01). The pipeline standardizes all timestamps using Pandas' mixed-format parser:

```
df["merger_cutoff_date"] = pd.to_datetime(df["merger_cutoff_date"], format="mixed").dt.strftime("%Y-%m-%d")
```

IV. DATABASE GATEWAY & RELATIONAL SCHEMA DESIGN

A. SQLAlchemy Engine Connection Management

Database credentials containing special characters (such as "@", "#", or "\$") cause URI connection parse errors if passed directly into database drivers. The pipeline uses RFC 3986 URL encoding via Python's `urllib.parse.quote_plus` to ensure reliable connection handshakes:

```
def get_engine(database=None): db_part = f'/{database}' if database else '' return
create_engine(f'mysql+pymysql://{DB_USER}:{DB_PASS}@{DB_HOST}:{DB_PORT}{db_part}')
```

B. Optimized Analytical SQL Aggregation

Rather than pulling unaggregated transactional records into memory on client machines, computational aggregation is pushed down into the MySQL engine. The following SQL query categorizes merger phases and calculates operational performance indicators:

```
SELECT CASE WHEN fiscal_year <= 2017 THEN 'Pre-Merger' ELSE 'Post-Merger' END AS
merger_phase, COUNT(*) AS branch_observations, ROUND(AVG(gross_npa_pct), 2) AS
avg_gross_npa_pct, ROUND(AVG(net_npa_pct), 2) AS avg_net_npa_pct, ROUND(AVG(roa_pct), 2) AS
avg_roa_pct, ROUND(AVG(cost_to_income_pct), 2) AS avg_cost_to_income_pct,
ROUND(AVG(car_crar_pct), 2) AS avg_capital_adequacy_pct, ROUND(AVG(casa_ratio_pct), 2) AS
avg_casa_ratio_pct, ROUND(SUM(advances_inr_cr) / 100000, 2) AS total_advances_lakh_cr,
ROUND(SUM(deposits_inr_cr) / 100000, 2) AS total_deposits_lakh_cr FROM sbi_branch_metrics GROUP
BY merger_phase ORDER BY merger_phase DESC;
```

V. BUSINESS INTELLIGENCE & ANALYTICAL MODELING

A. Power BI Semantic Model Structure

The processed dataset is ingested into Microsoft Power BI (`merger_dashboard.pbix`), where a specialized dimensional model supports interactive multi-dimensional analysis. Table II outlines the central DAX metrics implemented in the reporting engine.

TABLE II: Central Business Intelligence Measures Formulated in Power BI

Analytical DAX Measure	Underlying Formulation	Dashboard Functionality
Pre-Merger Avg GNPA	CALCULATE(AVERAGE(GNPA), Period="Pre-Merger")	Baseline risk benchmark for circle card comparisons
Post-Merger Avg GNPA	CALCULATE(AVERAGE(GNPA), Period="Post-Merger")	Tracks ongoing asset recovery post consolidation
GNPA Delta (pp)	[Pre-Merger Avg GNPA] - [Post-Merger Avg GNPA]	Heatmap KPI indicator across 12 administrative circles
Cost-Income Efficiency Gain	[Pre Cost/Income] - [Post Cost/Income]	Bar chart evaluating operational economies across branch tiers
Aggregate Advances (Lakh Cr)	DIVIDE(SUM(advances_inr_cr), 100000)	Monitors post-merger corporate & retail loan expansion

VI. PIPELINE PERFORMANCE EVALUATION & BENCHMARKS

A. Execution Latency and Throughput Benchmarks

To assess pipeline efficiency, we benchmarked each stage using execution profiling across cold and warm database connections. Table III details the latency and throughput profile of the system across standard test runs.

TABLE III: Pipeline Execution Latency and Computational Footprint

Execution Pipeline Stage	Mean Latency	Peak Memory Usage	Operational Output Metric
Stochastic Data Synthesis (generate_data.py)	0.42 sec	28.4 MB	4,000 synthetic records generated with dirty artifacts
Vectorized Sanitization & Imputation	0.18 sec	34.1 MB	100% missingness resolved; types coerced
Relational Database Bulk Ingestion	1.24 sec	42.5 MB	Full batch insert into MySQL (if_exists="replace")
Analytical Aggregation Queries (SQL)	0.04 sec	Server Cache	Full group-by aggregation across 8 fiscal years
Power BI Model Refresh & VertiPaq Load	1.85 sec	62.0 MB	Columnar compression and visual render

B. Architectural Scalability & Cloud Migration Path

While the current implementation runs locally on a MySQL and Python stack, its modular design enables straightforward cloud migration. For enterprise workloads exceeding 10^7 records, the pipeline can be modernized with minimal refactoring:

- **Distributed Processing:** Migrating data transformation from Pandas to Apache Spark (PySpark) allows distributed cluster scaling without altering the cleaning logic.
- **Cloud Data Warehousing:** SQLAlchemy's database dialect abstraction layer allows swapping MySQL for Snowflake, AWS Redshift, or Google BigQuery by modifying only the connection string.
- **Orchestration:** Production deployments can schedule the pipeline via Apache Airflow or Prefect to run incremental daily loads for ongoing branch surveillance.

VII. CONCLUSION

This paper presented the software architecture and engineering principles behind an end-to-end banking analytics pipeline. By combining algorithmic data generation, robust string and numerical cleaning, partition-based median imputation, and declarative SQL analytical queries, the architecture provides a reliable, reproducible foundation for analyzing large-scale institutional bank consolidations.

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